

	1	2	3	4	5	6	7	8
A	EC01 SENSOR UNIVERSAL Esquematico nativo KiCad REV 0.2 04 outubro 2026 Duplo clique em cada folha para abrir o circuito. Redes com o mesmo nome sao globais.							
B	Alimentacao e bateria Arquivo: Alimentacao.kicad_sch		ESP32 e Inicializacao Arquivo: ESP32.kicad_sch		BLE e cristal Arquivo: BLE_e_cristal.kicad_sch			
C	Radio LoRa 915 MHz Arquivo: LoRa.kicad_sch		Expansao e display Arquivo: Expansao_e_display.kicad_sch		Sensores externos e CAN Arquivo: Sensores_e_CAN.kicad_sch			
D	USB de servico Arquivo: USB_servico.kicad_sch							
E	BASE: 1S protegida -> buck-boost 3V3 -> ESP32-C3FH4 + E22-900M22S + sensores. RJ1/RJ2 sao um barramento comum. CAN exige alteracao de montagem; nao basta firmware. DNP = nao montar na base OneWire. Conectores e TVS TBD exigem selecao antes da PCB. Simulacoes PSpice e testes fisicos cancelados por solicitacao do cliente. Este projeto nao inclui PCB, antena final, firmware ou liberacao para fabricacao. Consultar EC01_Documento_de_Engenharia.pdf e BOM para decisoes, limites e pendencias.							
F	1	2	3	4	5	6	7	8

Footprints e partes TBD ainda nao liberados

ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /

File: EC01.kicad_sch

Title: EC01 Sensor Universal – Arquitetura e navegacao

Size: A3

Date: 2026-10-04

Rev: 0.2

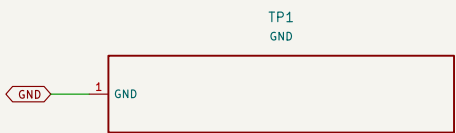
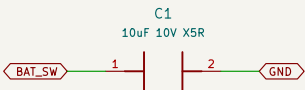
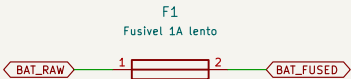
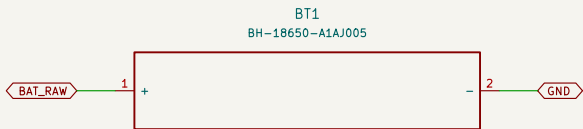
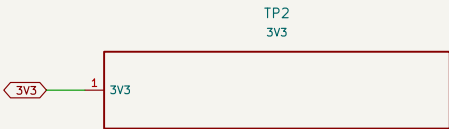
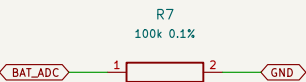
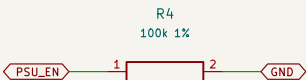
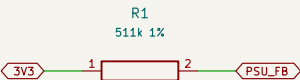
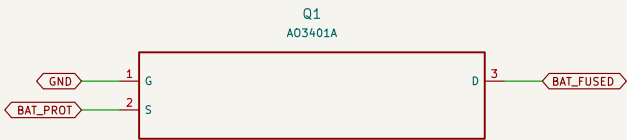
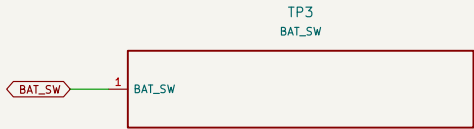
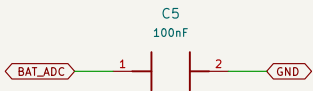
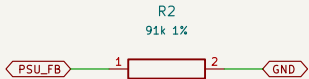
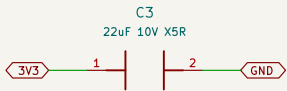
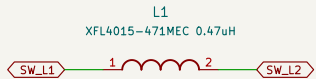
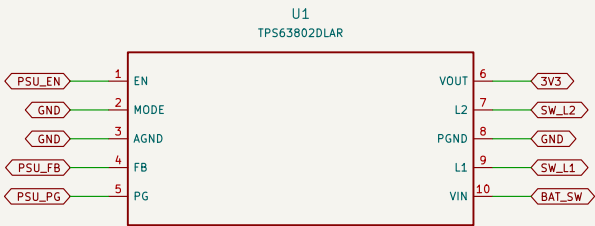
KiCad E.D.A. 10.0.6

Id: 1/8

Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: / File: EC01.kicad_sch		
Title: EC01 Sensor Universal – Arquitetura e navegacao		
Size: A3	Date: 2026-10-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 1/8

01 ALIMENTACAO E BATERIA



Celula 1S protegida e removivel. Sem carregador. USB nao alimenta o circuito.

TPS63802: 3,308 V nominal. EN: liga em -3,30 V; desliga em -3,00 V.

F1, F2, suporte e chave exigem selecao mecanica e coordenacao de protecao.



Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /Alimentacao e bateria/
File: Alimentacao.kicad_sch

Title: EC01 Sensor Universal – Alimentacao e bateria

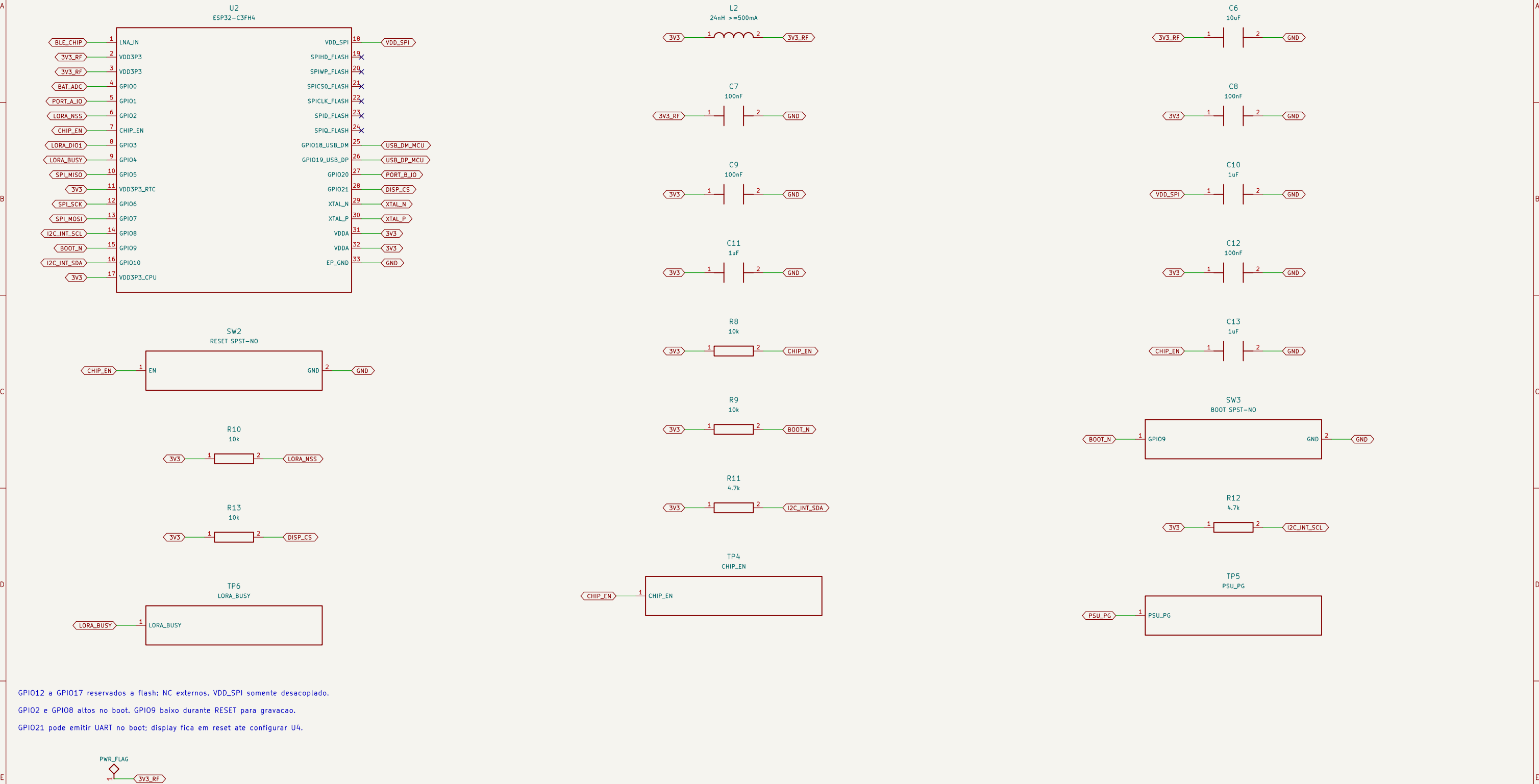
Size: A3 Date: 2026-10-04

Rev: 0.2

KiCad E.D.A. 10.0.6

Id: 2/8

02 ESP32 E INICIALIZACAO



GPIO12 a GPIO17 reservados a flash; NC externos. VDD_SPI somente desacoplado.
GPIO2 e GPIO8 altos no boot. GPIO9 baixo durante RESET para gravacao.
GPIO21 pode emitir UART no boot; display fica em reset ate configurar U4.

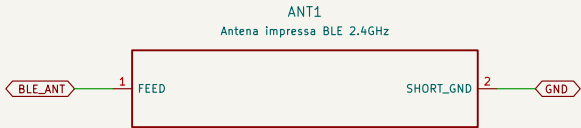
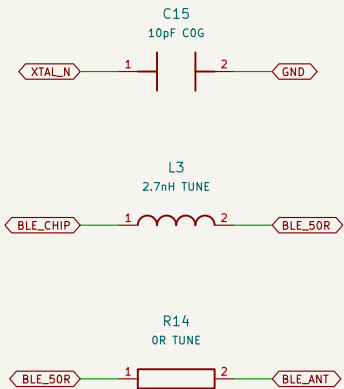
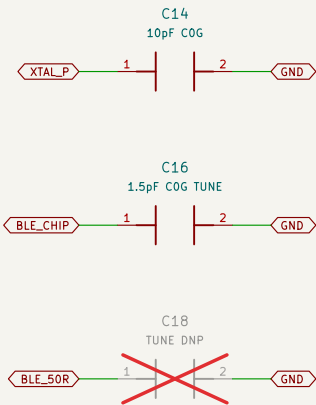
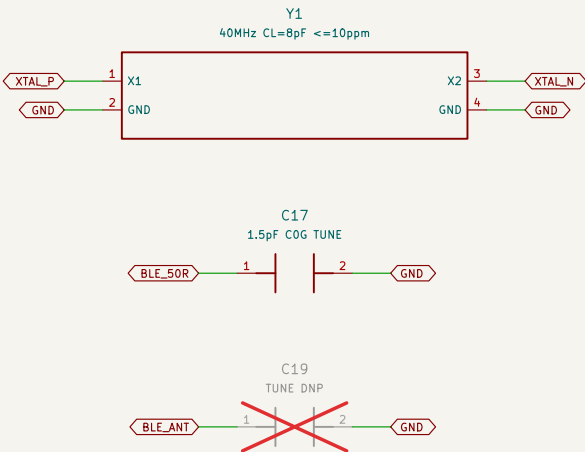
Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /ESP32 e inicializacao/
File: ESP32.kicad_sch

Title: EC01 Sensor Universal – ESP32 e inicializacao

Size: A3	Date: 2026-10-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 3/8

03 BLE E CRISTAL



LNA_IN isolado de 3V3. C16/L3/C17 sao valores iniciais para ajuste de RF.
ANT1 representa a IFA futura: geometria e casamento dependem da PCB e carcaca.
Cristal 40 MHz CL=8 pF: C14/C15=10 pF pressupoe -3 pF de parasitas.

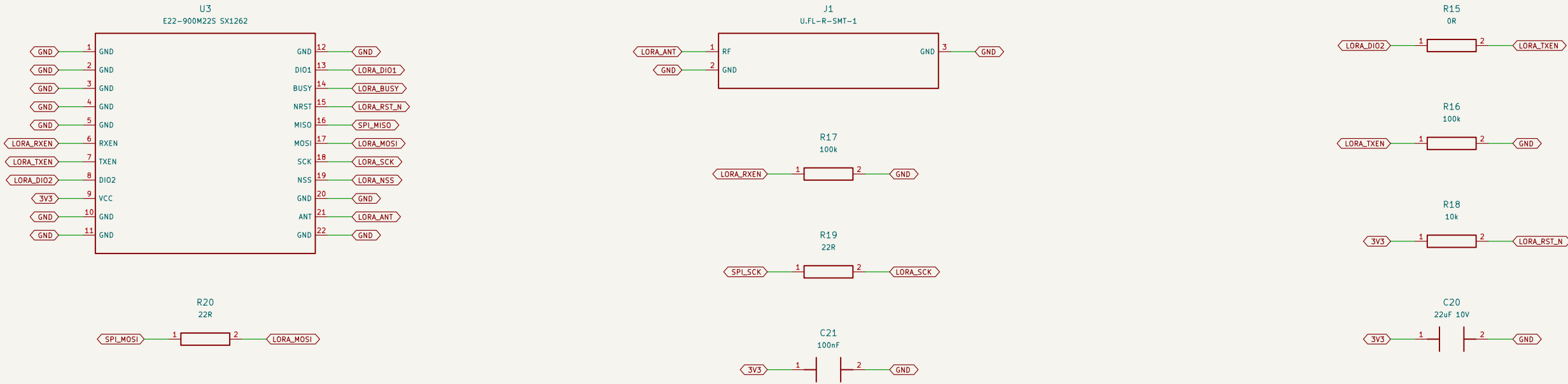
Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /BLE e cristal/
File: BLE_e_cristal.kicad_sch

Title: **EC01 Sensor Universal – BLE e cristal**

Size: A3	Date: 2026-10-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 4/8

04 RADIO LORA 915 MHZ

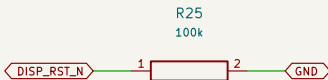
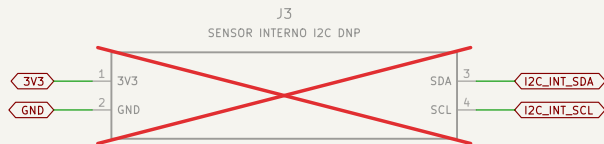
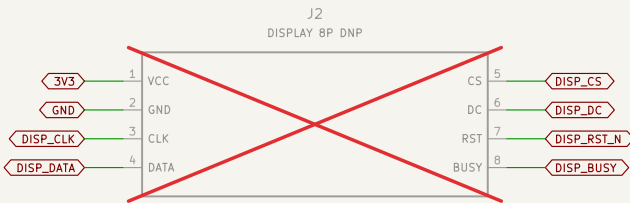
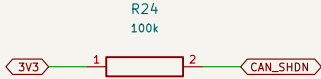


E22-900M225 sempre montado. Manual 2026: p12=GND, p15=NRST.
Firmware: DIO2 controla TXEN; TCXO interno por DIO3 a 2.2 V. RXEN via U4.
Usar somente J1 ou o IPEX integrado; evitar derivacao RF em aberto na PCB.

Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /Radio LoRa 915 MHz/ File: LoRa.kicad_sch		
Title: EC01 Sensor Universal – Radio LoRa 915 MHz		
Size: A3	Date: 2026-10-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 5/8

05 EXPANSAO E DISPLAY



U4 TCA9534A; endereco I2C de 7 bits 0x38. Inicializa com portas em entrada.

Display SPI: R28/R29. Display I2C: R30/R31. Nunca montar ambos os pares.

J2/J3 DNP. Interface 3.3 V para MODULO; nao aciona painel e--paper diretamente.

Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

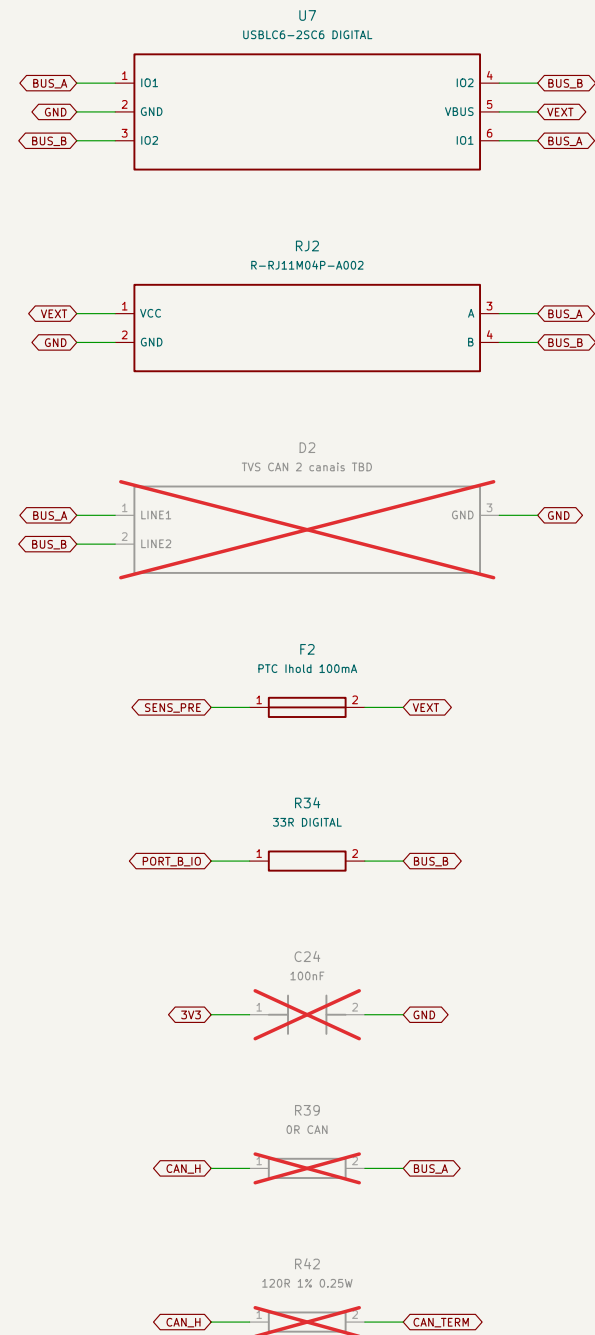
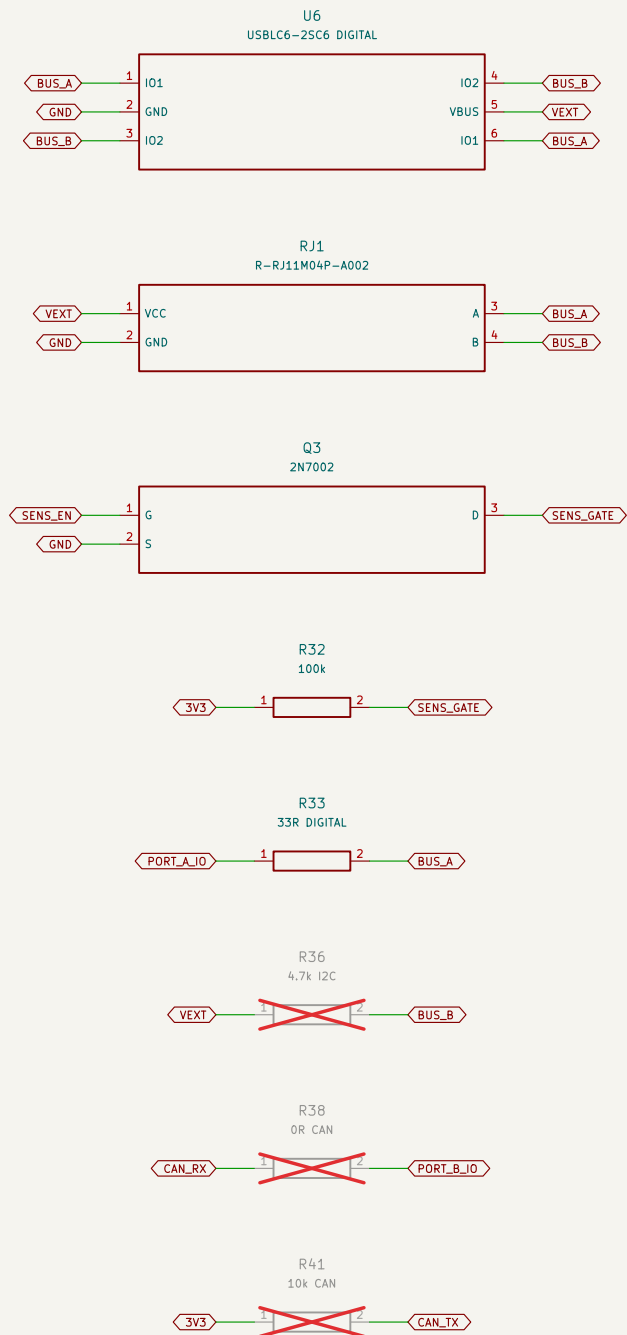
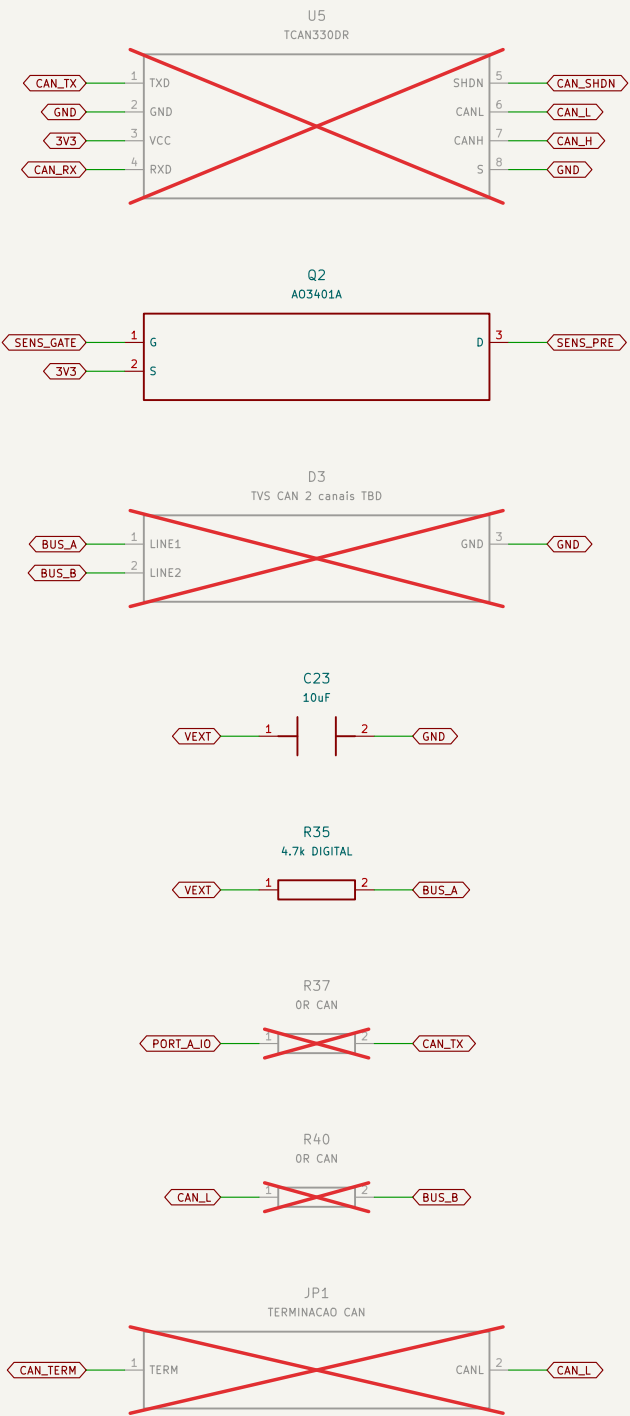
Sheet: /Expansao e display/
File: Expansao_e_display.kicad_sch

Title: EC01 Sensor Universal – Expansao e display

Size: A3 Date: 2026-10-04
KiCad E.D.A. 10.0.6

Rev: 0.2
Id: 6/8

06 SENSORES EXTERNOS E CAN



RJ1/RJ2 compartilham barramento. Pinagem LOGICA 1=VEXT 2=GND 3=A 4=B.

Base OneWire: R33/R34/R35 e U6/U7. I2C: acrescentar R36. Cabos a definir.

CAN: remover R33-R36 e U6/U7; montar U5/C24/R37-R41. TVS D2/D3 pendentes.

CAN usa um TWA1 comum. 120R por R42/JP1 somente na extremidade fisica.

VEXT 3,3 V nominal; orcamento 100 mA total. PTC nao e limitador preciso.

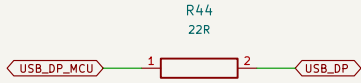
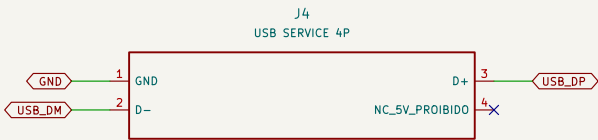
Com VEXT desligada, colocar GPIO1/20 em alta impedancia para evitar realimentacao.

Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /Sensores externos e CAN/
File: Sensores_e_CAN.kicad_sch
Title: EC01 Sensor Universal – Sensores externos e CAN

Size: A3 Date: 2026-10-04 Rev: 0.2
KiCad E.D.A. 10.0.6 Id: 7/8

07 USB DE SERVICIO



J4 e header de servico, nao USB-C. Pino 4 NC. Nunca conectar 5 V a 3V3.

Gravar com bateria ligada. D-/D+ por R43/R44 e ESD U8.

USB Serial/JTAG nativo: GPIO18 D- e GPIO19 D+. Verificar cabo e polaridade.

Footprints e partes TBD ainda nao liberados
ESQUEMATICO PRELIMINAR – SEM SIMULACAO – SEM PCB

Sheet: /USB de servico/
File: USB_servico.kicad_sch

Title: EC01 Sensor Universal – USB de servico

Size: A3	Date: 2026-10-04	Rev: 0.2
KiCad E.D.A. 10.0.6	Id: 8/8	